Chapter 06: Scoring, term weighting, and the vector space model

https://drive.google.com/file/d/1kP3AoSr6SnO2OqhpbP6TXNV4WNh1wxy5/view

📦 Bag of Words (BoW) Model

- document is treated as:
    - Just a **collection (multiset) of word**
    - **Order does NOT matter**
    - **Grammar does NOT matter**
    - **Frequency (how many times a word appears) DOES matter**

- Example

  D1: {John likes ~~to~~ watch movies. Mary likes movies ~~too~~. }

  D2: {John ~~also~~ likes ~~to~~ watch football games.}

  D3: { Mary ~~does~~ ~~not~~ like football games }

- Bow Representation

  D1: {John, like,watch,movie,Mary,like,movie}

  D2: {John,like,watch,football,game}

  D3: { Mary,like,football,game }

[^ After Preprocessing (lowercase, remove stopwords, stemming)]

- Query Term

  Q: {like football game}

📊 Similarity Calculation (Jaccard Similarity)

Formula used here:

$$\text{Similarity} = \frac{|D \cap Q|}{|D \cup Q|}$$

= (Common words) / (Total unique words)

1️⃣ (D1, Q)

D1: {John, like, watch, movie, Mary, like, movie}

Unique words in D1 = {John, like, watch, movie, Mary}

Q: {like, football, game}

🔹 Intersection (Common words)

= {like}

Count = 1

🔹 Union (All unique words)

= {John, like, watch, movie, Mary, football, game}

Count = 8

$$Similarity = 1/8$$

2️⃣ (D2, Q)

D2: {John, like, watch, football, game}

🔹 Intersection

= {like, football, game}

Count = 3

🔹 Union

= {John, like, watch, football, game}

Count = 5

$$Similarity = 3/5$$

3️⃣ (D3, Q)

D3: {Mary, like, football, game}

🔹 Intersection

= {like, football, game}

Count = 3

🔹 Union

= {Mary, like, football, game}

Count = 4

$$Similarity = 3/4$$

📌 Ranking Based on Similarity

Document	Score
D3	3/4 (Highest)
D2	3/5
D1	1/8 (Lowest)

So **D3 is most relevant** to the query.

⚠️ Problems with Bag of Words

Document size affects the overall similarity.

Frequency of common terms are ignored.

Similarity is only based on common terms, which are treated as independent of each other.

Vector Space Model is the answer to these problems.

Vector Space Model (VSM)

Vector Space Model is an **algebraic model** used to represent:

- Documents
- Queries

as **vectors in a multidimensional space**.

Each dimension = one term in the vocabulary.

Assumptions of Vector Space Model

✅ 1. Documents and queries are vectors

Both are represented in the same V-dimensional space.

✅ 2. Relevant documents are "close" to the query

If a document is relevant, its vector will be:

- Near the query vector
- Small angle between them

✅ 3. All terms are independent

Each term is treated separately.

Example:
"football" and "game" are independent.

(This is called the **independence assumption**.)

✅ 4. Term weighting matters

Not all words are equally important.

We assign weights using:

- TF (Term Frequency)
- IDF (Inverse Document Frequency)
- TF-IDF

Vector Representation of Documents

Step 1: Build Dictionary

From all documents, build a vocabulary.

Let vocabulary size = V

Example:

Vocabulary =

{ football, game, john, like, mary, movie, watch }

V = 7

Step 2: Represent Document as V-dimensional vector

Each document becomes:

$$d = (w_1, w_2, w_3, ..., w_V)$$

Each position corresponds to a word in dictionary (alphabetical order).

Example

Dictionary (alphabetical):

{ football, game, john, like, mary, movie, watch }

Now represent:

D2: "John likes to watch football games"

Vector:

football	game	john	like	mary	movie	watch
1	1	1	1	0	0	1

So:

$$d_2 = (1, 1, 1, 1, 0, 0, 1)$$

4️⃣ Vector Representation of Query

Query is represented the SAME way.

Q = { like, football, game }

Vector:

football	game	john	like	mary	movie	watch
1	1	0	1	0	0	0

So:

$$q = (1, 1, 0, 1, 0, 0, 0)$$

Only terms present in query have non-zero values.

5️⃣ Cosine Similarity

🔹 Why not Boolean Model?

Boolean model:

- Only gives match / no match
- No ranking

Vector model:

- Gives degree of similarity
- Allows ranking

Cosine Similarity Formula

$$\cos\theta = \frac{d \cdot q}{|d|\,|q|}$$

Where:

- $d \cdot q$ = dot product
- $|d|$ = magnitude of document vector
- $|q|$ = magnitude of query vector

Step-by-Step Example

Let:

$$d = (1, 1, 1, 1, 0, 0, 1)$$

$$q = (1, 1, 0, 1, 0, 0, 0)$$

Step 1: Dot Product

Multiply corresponding values:

$$(1 \times 1) + (1 \times 1) + (1 \times 0) + (1 \times 1) + (0 \times 0) + (0 \times 0) + (1 \times 0)$$

$$= 1 + 1 + 0 + 1 + 0 + 0 + 0 = 3$$

Step 2: Magnitudes

$$|d| = \sqrt{1^2 + 1^2 + 1^2 + 1^2 + 0 + 0 + 1^2}$$

$$= \sqrt{5}$$

$$|q| = \sqrt{1^2 + 1^2 + 0 + 1^2 + 0 + 0 + 0}$$

$$= \sqrt{3}$$

Step 3: Cosine Similarity

$$\cos\theta = \frac{3}{\sqrt{5}\sqrt{3}}$$

$$= \frac{3}{\sqrt{15}}$$

[iske aagey ka sir ne class mein toh nhi karaya tha BUTT its mentioned in syllabus so idk mein nhi krrhi lekin goodnight] !!!!!!!!!!!!!!!!!!!!!!!!!!